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Solar PV Fault Localization and Rectification

Group 22 Literature Review by

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Introduction

Over the years, global renewable energy has played a major role in combating climate change thus the rapid expansion of solar photovoltaic (PV) systems has introduced new challenges in maintaining operational reliability and energy efficiency. As PV installations expand to utility-scale farms, post-installation monitoring and maintenance have become crucial due to their exposure to environmental, electrical, and mechanical stress. Delays in identifying faults in these farms can lead to major energy yield losses, prolonged downtime, as fault localisation is often performed manually. This, in turn, results in consequent financial losses. To address these challenges, we propose an integrated solution that combines fault detection, localisation at the string and cell levels, fault severity assessment, and fault rectification. This system aims to enhance system reliability, reduce diagnostic time, and minimize downtime, thereby improving the profitability of solar PV operational efficiency and profitability of solar PV installations whilst improving its operational efficiency.

Previous Work

Fault Detection

Fault detection in photovoltaic (PV) systems is highly important in maintaining system efficiency and preventing failures that can lead to energy loss.

Hajji et al. (2021) adopted an approach which used multivariate feature extraction with supervised learning, where principal component analysis (PCA) was used to reduce data dimensionality and extract important features such as T^2, SPE and SWE. Afterwards, they were fed to classifiers such as K-Nearest Neighbours (KNN), Random Forest (RF), and Support Vector Machines (SVM). When testing with real PV system data, the method showed to have a high accuracy, low false alarms and effective fault discrimination. While effective, this approach is limited in adaptability for complex fault patterns due to manually extracted features.

Zaki et al. (2020) further enhanced this approach using deep learning (DL) to enable automatic feature extraction. They combined Convolutional Neural Networks (CNN) with meta-heuristic modelling, an Improved Teaching-Learning-Based Optimisation (ITLBO) to accurately model the system. By using normalised electrical indicators, including voltage, current, and fill factor, the model extracted features and classified them into one healthy and six fault types, achieving over 98% accuracy and outperforming standalone classifiers such as SVM, ANN, and PNN. However, this DL framework results in higher computational costs.

Lodhi et al. (2022) used an AdaBoost ensemble (AEM) algorithm to improve the classification robustness and fault detection accuracy. SVM was used as a weak learner, and they evaluated the AEM on a simulated PV string and on measured signals. This method detected open-circuit, closed-circuit, and degradation faults using multiple electrical parameters, achieving 97.84% accuracy and outperforming standalone classifiers such as SVM, RF, KNN. Although it has a slightly lower accuracy than Hajji, et al. (2021), the ensemble approach reduces overfitting and provides better robustness by combining multiple weak models.

Building on feature-based and indicator-based approaches, Li et al. (2021) proposed a more comprehensive approach by analysing a full I-V curve analysis of PV systems. They simulated a six-module (with 50W each) array that had eight conditions (healthy plus seven faults) with varying irradiance and temperature. Machine learning (ML) classifiers including Artificial Neural Networks (ANN), SVM, Decision Trees (DT), RF, KNN, Naïve Bayes (NB) were tested with PCA for dimensionality reduction. The ANN with GADF achieved an accuracy of 100% on both simulated and experimental data, and was faster after PCA. This method focuses on complete curve representations than electrical indicators, offering better fault information, but requires intensive pre-processing before classification.

Finally, Sridharan et al. (2023) explored a visual fault detection approach using DL and weightless neural networks (WNN). By capturing RGB images of PV modules via UAVs, features were extracted using DenseNet-201, and the impactful features were selected using a J48 decision tree. With this, a WiSARD classifier was used for fault classification, which achieved an accuracy of 100% while maintaining low computational cost, hence it is suitable for use in real-time.

After Identifying effective methods for fault detection, the next logical step is determining where the fault occurs. This Introduces the concept of fault localisation.

Fault Localization

Localisation is the identification of the exact position of faults in cells, individual modules, strings, or PV arrays within a photovoltaic system. Despite a fault being detected, manual inspection is required to identify the root cause, making the process labour-intensive and time-consuming. Therefore, using ML and DL algorithms in a localisation model can map the location of faults, minimising downtime and operational cost in large PV farms.

Venkatakrishnan et al. (2023) provided a comprehensive review of different approaches used for fault detection and localisation in photovoltaic (PV) systems. This review showed the research in this field remains fragmented across different modalities, such as electrical signals, sensor arrays, and imagery etc. They found that integration techniques for electrical and image-based data would further improve fault classification. However, the accuracy of these methods still depends mostly on the data quality and the environmental conditions under which the system is tested.

Building on this emphasis on image-based diagnostics, Ahan et al. (2021) applied DL techniques to EL imagery for finer cell-level localisation. They compared several DL architectures, including EfficientNet-B7, Xception, Inception variants, DenseNet, ResNeXt, as well as detection frameworks like RetinaNet and Faster R-CNN. The experiments were conducted on both monocrystalline and polycrystalline PV cells, achieving 94% accuracy through DL CNNs, demonstrating the effectiveness of image- based analysis. The study found that combining modern convolutional architectures with suitable detection models effectively locates subtle defects in a range of PV materials.

While Ahan et al. focused on visual data from EL images, Chen and Wang (2017) shifted their focus to temporal electrical signals. They investigated adaptive fault localisation using simulated PV arrays in PSCAD/EMTDC and explored both Random Forest and sequence-model architectures (1D-CNN, LSTM, Bi-LSTM). The results suggest that when time-series characteristics of PV string behavior are captured in simulation, recurrent and convolutional temporal models can effectively distinguish fault signatures, indicating that this sequence model can exploit temporal dependencies in electrical variables, assisting in image-based methods.

In contrast to these sequence models, Zhi-cong Chen et al. (2018) demonstrated that the traditional ensemble approach can also deliver outstanding results. Their well-designed Random Forest model, using simulated array voltage and string current measurements, achieved 99.9% accuracy. Proving data quality and feature selection matter as much as the model's complexity. Overall, while DL models achieve higher accuracy through automated feature extraction, ensemble methods such as Random Forests offer simpler implementation, faster computation, and strong performance with lower data and computational requirements. After pinpointing the location of the fault, it is essential to assess its severity is the next crucial step, aiding in planning out an effective maintenance response.

Fault Severity

Solar photovoltaic fault severity and forecasting remain underexplored despite their importance in maintaining power stability, minimizing losses, and enabling timely maintenance before faults escalate. This section synthesizes key studies, comparing their contributions and limitations to outline an integrated framework for assessing both the occurrence and severity of PV faults.

Farooq et al. (2025) employed Gradient Boosting Machines and Linear Regression to forecast power and detect anomalies from inverter and sensor level data. GBM achieved strong accuracy (0.97 and 0.61), demonstrating that monitoring temporal power fluctuations can effectively indicate fault progression. However, its short testing period of 34 days limits its long term applicability. Our work extends this premise by building on their anomaly based methodology while aiming for improved temporal depth and model reliability.

While Farooq et al. focused on power fluctuation driven faults, degradation based studies provide deeper insights into fault evolution. Khan et al. (2024) used CNNs, SVMs, and Random Forests to classify fault severity from imaging and electrical data, achieving up to 99% accuracy. This demonstrates that degradation signatures, both electrical and visual, serve as consistent indicators of severity and fault type, enabling early stage severity estimation.

Similarly, Basem et al. (2025) combined Numerical Modelling with Emotional Artificial Neural Networks to simulate degradation under limited and nonlinear data conditions. Their approach addressed the challenges of incomplete datasets, but its reliance on ideal weather conditions restricted environmental realism. Integrating this modelling strength with variable climatic data, as our research intends, could significantly enhance real-world severity prediction.

Environmental variability remains a parallel driver of fault behavior. Tripathi et al. (2024) compared Support Vector Machine Regression, Multivariate Regression, and Gaussian Process Regression, which were the most accurate ($R^2 = 0.99$, MAE = 0.17). Their work reinforces that accurate modelling of expected power under changing conditions allows deviations to act as quantitative indicators of fault severity.

Dhingra et al. (2022) further explored degradation-focused modelling using ensemble regressors such as Extra Tree, Random Forest, and CatBoost to predict degradation rates from a global climate dataset, achieving 100% accuracy with ETR. These findings emphasize the importance of identifying which regression models most effectively capture the interaction between degradation trends and environmental fluctuations. However, even high-performing regressors exhibit limitations in handling long term degradation uncertainty and extreme environmental variability. Our research aims to refine these models, improving their adaptability and accuracy across diverse conditions to advance the precision of fault severity prediction and early warning mechanisms in PV systems.

After evaluating fault severity and predicting potential fault behaviours, the next essential step would be applying these insights to the final phase called fault rectification.

Fault Rectification

Fault rectification in photovoltaic (PV) arrays has been significantly enhanced through the application of machine learning, enabling precise detection, diagnosis, and optimization to ensure reliable solar energy operations

For instance, "Machine Learning for Predictive Maintenance in Solar Farms" by Qureshi et al.(2024) presents a comprehensive analysis of leveraging machine learning techniques to forecast equipment failures and optimise maintenance in solar energy infrastructure. Emphasising proactive strategies over reactive ones, the research highlights data acquisition, feature engineering, model training, and deployment as key components, addressing challenges like sensor data heterogeneity and real-time monitoring. Common ML algorithms discussed include logistic regression, decision trees, SVM, clustering, and anomaly detection for enhanced reliability and efficiency. By minimising downtime and costs, the approach maximises energy production, drawing on real-world data to inform scalable solutions in renewable energy operations (Qureshi et al., 2024).

Building on the theme of improving solar energy system performance, “Optimization of Solar Panel Deployment using machine learning” by Kamal et al. (2022) proposes a machine learning-assisted mechanism for reconfiguring photovoltaic array topologies, encompassing series parallel, parallel, bridge link, honeycomb, and total cross-tied configurations. Utilising artificial neural networks, the method optimises electrical connections under partial shading by training on synthetic irradiance profiles to predict optimal topology, thereby maximising power output with additional hardware. Simulations reveal superior performance compared to support vector machines, Naive Bayes, and k-nearest neighbour algorithms, attaining an average test accuracy of 95% with precision up to 0.8941, recall up to 0.956,4, and f-measure up to 0.9412, enhancing PV efficiency across varied conditions (Kamal et al., 2022).

Building on these optimization strategies, Abdulla et al. (2024) explored techniques to enhance PV system reliability through fault rectification and predictive maintenance. Faults in panels and inverters are efficiently detected using infrared thermography, SCADA monitoring, and machine learning. Predictive maintenance uses anomaly detection and historical data to forecast equipment deterioration, while AI models like neural networks and LSTM help anticipate failures, improving system reliability and cost-efficiency.

Similarly, advancements in forecasting accuracy further enhance the operational efficiency of the PV systems. In photovoltaic (PV) power forecasting, the hybrid deep learning model improves accuracy by processing residuals generated by a non-pooling convolutional neural network (NPCNN). Utilising wallet transform(WT), it decomposes raw forecasting errors into sub-frequency sequences, separating high-frequency uncertainties from input data and low-frequency over-fitting artefacts. K-nearest neighbour(KNN) then extracts nonlinear features from each sequence for error prediction. Validation dataset errors inform the process, enabling reconstruction of refined forecasts. Evaluated on Limburg, Belgium PV data, this WT+KNN ECM reduces biases, enhancing ultra-short term accuracy over benchmarks like CNN and LSTM, as simulations demonstrate superior performance (Zhan et al., 2022)

Further extending the application of machine learning, the fault diagnosis method in Mellit et al. 's (2021) study," A Machine Learning and Internet of Things-Based Online Fault Diagnosis Method for Photovoltaic Arrays", utilises a decision tree for detection and a random forest for classification of anomalies in PV systems. Implemented on a Raspberry Pi 4 with IoT integration, it processes I-V curve features alongside solar irradiance and temperature data. Tested on a small-scale PV array in Jijel, Algeria, the approach identifies faults such as soiling, permanent shading, module disconnection, and short-circuited bypass diodes with 98% detection and 96% classification accuracy, facilitating timely rectification and outperforming traditional techniques in real-time applications. Together, these studies demonstrates the evolution of PV fault management system with the integration of detection, localisation, severity evaluation and rectification.

Comparing Previous Work

Research	Author	Year	Dataset	Model	Metric
Fault Detection					
Multivariate feature extraction based supervised machine learning for fault detection and diagnosis in photovoltaic systems	Hajji, M. et al.	2021	Collected from an emulator of a GCPV system	KNN, RF, SVM, DA, NB, DT	Train and test accuracies: RF – 99.64%, 99.87% KNN – 99.07%, 99.43% SVM – 98.66%, 99.18% DT – 97.24%, 97.29% NB – 93.32%, 93.21% DA – 75.22%, 75.75%
Fault diagnosis of photovoltaic panels using full I–V characteristics and	Li, B. et al.	2021	Simulated: 12K train, 2.4K test, 8 conditions (7 faults+healthy), 120 experimental curves	SVM, ANN, DT, RF, KNN	8paras: SVM – 94.83% Direct I-V: ANN - 99.92% RP: ANN – 99.96% GADF: ANN – 100%

	.				PCA - 99.92%,99.96%, 100% No PCA- 99.88%,99.92%,100%
Deep-learning–based method for faults classification of PV system	Zaki, S.A. et al.	2020	Simulation and experimental data from a 9.56kW PV array	1D CNN (RMSProp optimizer)	Simulation: 98.3-98.9% Experimental: 96.76-97.41%
An AdaBoost Ensemble Model for Fault Detection and Classification in Photovoltaic Arrays	Lodhi, E. et al.	2022	2000 simulated samples, 500 for each case	AEM, SVM, RF, KNN	AEM - 97.84% SVM - 96.76% RF - 94.34% KNN - 91.29%
Weightless Neural Network-Based Detection and Diagnosis of Visual Faults in Photovoltaic Modules	Sridharan et al.	2023	Acquired using an unmanned aerial vehicle	DenseNet-201, J48 Decision Tree, WiSARD	Accuracy - 100%
Fault Localization					
Detection, location, and diagnosis of different faults in large solar PV system—a review	G.R. Venkatakrishnan, R. Rengaraj, S. Tamilselvi, J. Harshini, Ansheela Sahoo, C. Ahamed Saleel, Mohamed Abbas, Erdem Cuce, C. Jazlyn, Saboor Shaik, Pinar Mert Cuce, Saffa Riffat	2023		KNN, EWMA, UEWMA, MEWMA, MSWGLTR, T-Test, KELM, ANN, PNN, MLP, DT, LAPART, KF, ANFIS, RBF-KELM, LOF, EDM, NARX, Fuzzy, K-Means, Silhouette, LIT, EL, PLA, CMM, TDR, SSTDR, FMEA, ARIMA, ESA, IV, ECM, Satellite, Visual, IR, Sensors	KNN – 98.7% ANN – 99.2% PNN – 98.6% DT – 99.0% LAPART – 98.3% KF – 97.5% ANFIS – 98.8% RBF-KELM – 99.1% LOF – 96.8% EDM – 95.6% NARX – 97.9% Fuzzy – 96.0% K-Means – 94.7% TDR – 95.0% IV – 96.8% Visual – 90.0% IR – 98.5% Sensors – 99.5%

AI-assisted Cell-Level Fault Detection and Localization in Solar PV	Ahan M R, Akshay Nambi, Tanuja Ganu, Dhananjay Nahata,	2021	dataset curated by ZAE Bayern, which contains both mono-crystalline and	EffNet-B7 + GAP, XcepNet, IncV3, IncResV2, DenseNet101,	EffNet-B7 + GAP – 94.24% XcepNet – 92.19% IncV3 – 87.58% IncResV2 –
Electroluminescence Images	Shivkumar Kalyanaraman		poly-crystalline EL Images	ResNeXt50, RetinaNet, FRCNN, KNN	84.71% DenseNet101 – 79.23% ResNeXt50 – 81.17% RetinaNet – 89.2% FRCNN – 87.2% RetinaNet – 88.7% FRCNN – 94.2%
Adaptive Fault Localization in Photovoltaic Systems	Leian Chen and Xiaodong Wang	2017	Simulated using PSCAD/EMTDC (photovoltaic array simulator)	RF, 1D CNN, LSTM, Bi-LSTM	RF - 99.9% 1D CNN - 99.9% LSTM - 99.9% Bi-LSTM - 99.94%
Deep Learning-Based Model for Defect Detection and Localization on Photovoltaic Panels	S.Prabhakaran, R. Annie Uthra, J. Preetharoselyn	2022	Real-time RGB images of PV panels, images are preprocessed by applying the RHA algorithm	RMVDM, HOLT, CNN, YOLOv3, ResNet50	Detection Accuracy - Localization Accuracy - 97% False Classification Ratio - 95% Time Complexity-high
Random forest based intelligent fault diagnosis for PV arrays using array voltage and string currents	Zhi-cong chen, fuchang han, lijun wu, jinling yu	2018	Simulink based accurate PV simulation system obtain simulation data samples under various ambient conditions	RF model	RF algorithm - 99.9%
Fault Localization in Photovoltaic Systems Using Machine Learning	Muhannad Muhammad Hamad,Abdolreza Sheikholeslami	2024	Relied on simulated data	SVM, DT, FNN, RF	SVM - 75% DT -82% FNN -85% RF - 90%
Fault Severity					

Time Series Analysis of Solar Power Generation Based on Machine Learning for Efficient Monitoring	Farooq. et al.	2025	Data was collected from two different solar power plants in India over 34 days, including power generation data at the inverter level and sensor data at the plant level.	GBC, LR	LR scored 0.97 and 0.62, while GBC scored 0.97 and 0.61 for Plant 1 and Plant 2, respectively
A comprehensive analysis of advanced solar panel productivity and efficiency through numerical models and emotional neural networks	Basem. et al.	2025	The study utilized a dataset collected from a 200 W solar panel (model F200P-24) over a full day (7 a.m. to 5 p.m.)	EANN, NM	EANN has a high accuracy with lower RMSE compared to numerical model. NM has a moderate accuracy, higher RMSE than EANN
Advancing solar PV panel power prediction: A comparative machine learning approach in fluctuating environmental conditions	Tripathi et al.	2024	The dataset was collected through experiments at a solar installation with two 3.2 kW PV stations.	MR, SVMR, GPR	SVMR performed best with R^2 = 0.99, MAE = 0.17, and MSE = 0.038, followed by MR (0.93, 0.39, 0.28) and GPR (0.88, 0.63, 0.49).
A Review of Degradation and Reliability Analysis of a Solar PV Module	Khan, Z.U., et al.	2024	Both real field and simulated Datasets are used for non-destructive imaging methods along with electrical parameters	SVM, DT, RF, KNN, PNN, CNNs.	CNN: 90–99%, SVM: 85–95%, RF: 83–92%, KNN & PNN: 80–90%, DT: 75–85%
A Comparative Study of Machine Learning Algorithms for Photovoltaic Degradation Rate Prediction	Dhingra, B. et al.	2022	The dataset used in the study contains global climate data related to PV module degradation	ETR, CBR, RFR, XGBoost, DTR, LGBM, LR, RR, KNR	Based on the Mean Absolute Error: ETR-100% RFR-75%

			mechanisms and rates, with 193,557 entries		CBR-46% DTRr-0.77% XGBoost-27.91% LGBM-23.08% LR-11.88% RR-11.65% KNR-6.59%
Fault Rectification					
Optimizing maintenance schedules and corrective actions in solar farms using ML based predictive insights	Muhammad Salik Qureshi, Shayan Umar, Muhammad Usman Nawaz	2024		Logistic Regression, Decision Trees, Random Forests, SVM, Gradient Boosting	Accuracy(94.2 - 96.5%)Accuracy(94.2 - 96.5%) Precision(95.8 %) Recall(97.2%) F1-Score(96.5%)
Rectification of Solar PV arrays using machine learning	Shoaib Kamal et al.	2022	Synthetic irradiance profiles for a 3x4 PV array, generated using binary mapping simulated in MATLAB/Simulink using the Scandia performance model for PV modules	ANN	Accuracy(95 - 98 %) Precision(0.89) Recall(0.95)
Photovoltaic Systems Operation and Maintenance	Hind Abdulla Andrei Sleptchenko Ammar Nayfeh	2024	186 articles from Scopus and Web of Science(2010 - 2023)	Bibliometric analysis, PRISMA framework, Prognostic models, Predictive maintenance models, Reinforcement learning, Decision support systems	
Rectification of PV power forecasts	Rongquan Zhang,	2022		Hybrid Model which uses IF,	

using WT and KNN for error correction	Gangqiang Li, Siqi Bu, Guiwen Kuang, Yuxiang Zhu, Saddam Aziz		Historical PV power data from Limburg,Belgium	Non-pooling CNN, WT and KNN error correction module	MAE MAPE RMSE
ML and IoT-based online system detects and classifies PV array faults to improve performance	Adel Mellit, Omar Herrak, Catalina Rus Casas, Alessandro	2021	Voltage and current measurements (I-V curves) collected from a small-scale PV system with three parallel connected PV modules at the Renewable Energy Laboratory, University of Jijel, Algeria	Decision Tree for fault detection, Random Forest for fault classification and Raspberry Pi 4 for implementation	Detection accuracy 98% Classification accuracy 96%

Summary

These related studies demonstrate that PV fault management has evolved beyond manual checks and has evolved to using intelligent, data-driven approaches. In detection, feature extraction, and ensemble learners such as Random Forests and AdaBoost have improved accuracy and reduced false alarms, while deep learning methods (CNNs combined with meta-heuristic tuning) enable automatic feature extraction. DenseNet-201 along with the WiSARD classifier had the highest accuracy using selected features and maintaining low computational cost via WNNs. In localisation, electroluminescence imagery, time-series electrical data, and mixed datasets have improved diagnostic accuracy, although most studies still focus on cell or module level mapping. Recent research proves that models such as CNN-SVMa, Gradient Boosting, and regression ensembles can predict fault severity under multiple environmental conditions.

Extending on these observations, our proposed system introduces an integrated fault management system consisting of four key components, which include fault detection, fault localisation, fault severity, and fault rectification. It incorporates string-level localisation in which precision is achieved through fault mapping, resulting in faster isolation of defect areas. The output from the localisation method is fed into the fault severity model, which identifies the severity and the effect on the system. Finally, the rectification model assists in providing predictive insights, suggesting optimal corrective measures to avoid the same issue. With this approach, the system aims to assist professionals in the PV field since it reduces the diagnostic time, improves maintenance accuracy, minimizes energy loss, and the cost-effectiveness of solar operations. By doing so, it effectively bridges the existing research gap, contributing to the future of self-sustaining renewable energy management systems.

References

1. Ahan, M.R., Nambi, A., Ganu, T., Nahata, D. and Kalyanaraman, S., 2021. 'AI-assisted cell-level fault detection and localisation in solar PV electroluminescence images.' *Conference on Embedded Networked Sensor Systems*. Coimbra (Portugal), 15-17 November. New York, NY: Association for Computing Machinery. pp. 485-491. Available at: https://doi.org/10.1145/3485730.3493455

2. Abdulla, H., Andrei Sleptchenko and Ammar Nayfeh (2024). Photovoltaic systems operation and maintenance: A review and future directions. Renewable & sustainable energy reviews, 195, pp.114342–114342. https://doi.org/10.1016/j.rser.2024.114342
3. Basem, A., Opakhai, S., Elbarbary, Z.M.S., Atamurotov, F. and Benti, N.E., (2025). 'A comprehensive analysis of advanced solar panel productivity and efficiency through numerical models and emotional neural networks.' *Scientific Reports,* 15(1), p.259. Available at: https://doi.org/10.1038/s41598-024-70682-2
4. Chen, L. and Wang, X., (2017). 'Adaptive fault localisation in photovoltaic systems.' *IEEE Transactions on Smart Grid,* 9(6), pp. 6752–6763. Available at: https://doi.org/10.1109/TSG.2017.2722821
5. Chen, Z., Han, F., Wu, L., Yu, J., Cheng, S., Lin, P. and Chen, H., (2018). 'Random forest based intelligent fault diagnosis for PV arrays using array voltage and string currents.' *Energy Conversion and Management,* 178, pp. 250–264. Available at: https://doi.org/10.1016/j.enconman.2018.10.040
6. Dhingra, B., Tyagi, S. and Tomar, A., (2022). 'A comparative study of machine learning algorithms for photovoltaic degradation rate prediction.' *Conference on Intelligent Computing, Instrumentation and Control Technologies (ICICICT).* Kannur (India), 11-12 August. New York, NY: IEEE, pp. 474–478. Available at: https://doi.org/10.1109/ICICICT54557.2022.9917960
7. Farooq, U., Mushtaq, M.F., Ullah, Z., Ejaz, M.T., Akram, U. and Aslam, S., (2025). 'Time series analysis of solar power generation based on machine learning for efficient monitoring.' *Engineering Reports,* 7(2), e70023. Available at: https://doi.org/10.1002/eng2.70023
8. Hajji, M., Harkat, M.F., Kouadri, A., Abodayeh, K., Mansouri, M., Nounou, H. and Nounou, M., (2021). 'Multivariate feature extraction based supervised machine learning for fault detection and diagnosis in photovoltaic systems.' *European Journal of Control,* 59, pp. 313–321. Available at: https://doi.org/10.1016/j.ejcon.2020.03.004
9. Hamad, M.M. and Sheikholeslami, A., 2024. 'Fault localisation in photovoltaic systems using machine learning.' *Conference on Intelligent Systems (ICIS).* IEEE, pp.1–4. https://doi.org/10.1109/ICIS64839.2024.10887405
10. Kamal, S., Alzahrani, A., Olatomiwa, L., Alquthami, T., and Khan, M., 2022. Optimisation of solar panel deployment using machine learning. *International Journal of Photoenergy,* 2022, pp.1–7. https://doi.org/10.1155/2022/7249109
11. Khan, Z.U., Khan, A.D., Khan, K., Al Khatib, S.A.K., Khan, S., Khan, M.Q. and Ullah, A., 2024. A review of degradation and reliability analysis of a solar PV module. *IEEE Access,* 12, pp.185036–185056. https://doi.org/10.1109/ACCESS.2024.3432394
12. Li, B., Delpha, C., Migan-Dubois, A. and Diallo, D., 2021. Fault diagnosis of photovoltaic panels using full I–V characteristics and machine learning techniques. *Energy Conversion and Management,* 248, p.114785. https://doi.org/10.1016/j.enconman.2021.114785

13. Lodhi, E., Wang, F.Y., Xiong, G., Dilawar, A., Tamir, T.S. and Ali, H., 2022. An AdaBoost ensemble model for fault detection and classification in photovoltaic arrays. *IEEE Journal of Radio Frequency Identification,* 6, pp.794–800. https://doi.org/10.1109/JRFID.2022.3212310.
14. Mellit, A., Herrak, O., Rus Casas, C. & Massi Pavan, A., 2021. A machine learning and Internet of Things-based online fault diagnosis method for photovoltaic arrays. *Sustainability*, 13(23), p.13203. https://doi.org/10.3390/su132313203
15. Prabhakaran, S., Uthra, R.A. and Preetharoselyn, J., 2023. Deep learning-based model for defect detection and localisation on photovoltaic panels. *Computer Systems Science & Engineering,* 44(3).
16. Qureshi, M.S., Umar, S. and Nawaz, M.U. (2024). Machine Learning for Predictive Maintenance in Solar Farms Introduction. ResearchGate, [online] p.3. Available at: https://www.researchgate.net/publication/390178131_Machine_Learning_for_Predictive_Maintenance_in_Solar_Farms_Introduction [Accessed 05 October 2025].
17. Sridharan, N.V., Joseph, J.V., Vaithiyanathan, S. and Aghaei, M., 2023. Weightless neural network-based detection and diagnosis of visual faults in photovoltaic modules. *Energies,* 16(15), p.5824. https://doi.org/10.3390/en16155824
18. Tripathi, A.K., Aruna, M., Elumalai, P.V., Karthik, K. et al., 2024. Advancing solar PV panel power prediction: A comparative machine learning approach in fluctuating environmental conditions. *Case Studies in Thermal Engineering,* 59, 104459. https://doi.org/10.1016/j.csite.2024.104459
19. Venkatakrishnan, G.R., Rengaraj, R., Tamilselvi, S., Harshini, J., Sahoo, A., Saleel, C.A., Abbas, M., Cuce, E., Jazlyn, C., Shaik, S., Cuce, P.M. and Riffat, S., 2023. Detection, location, and diagnosis of different faults in large solar PV system—a review. *International Journal of Low-Carbon Technologies,* 18, pp.659–674. https://doi.org/10.1093/ijlct/ctad018
20. Zaki, S.A., Zhu, H., Fakih, M.A., Sayed, A.R. and Yao, J., 2021**.** Deep-learning-based method for fault classification of PV system. *IET Renewable Power Generation,* 15(1), pp.193–205. https://doi.org/10.1049/rpg2.12016
21. Zhang, R., Li, G., Bu, S., Kuang, G., He, W., Zhu, Y. and Aziz, S. (2022). A hybrid deep learning model with error correction for photovoltaic power forecasting. *Frontiers in Energy Research*, 10. https://doi.org/10.3389/fenrg.2022.948308